Networking Basics: DHCP

DHCP (Dynamic Host Configuration Protocol) is a network protocol that automatically assigns IP addresses and other network settings to devices on a network. This eliminates the need to configure each device manually.

Why DHCP is Used

- Automatically assigns IP addresses to devices.
- Reduces manual configuration errors.
- Saves time for network administrators.
- Makes it easy to manage large networks.

Information DHCP Provides

When a device connects to a network, DHCP can assign:

- **IP Address** – Unique address for the device.
- **Subnet Mask** – Defines the network and host portions of the IP address.
- **Default Gateway** – The router used to communicate outside the local network.
- **DNS Server Address** – Converts domain names (e.g., google.com) into IP addresses.
- **Lease Time** – How long the device can use the assigned IP address.

How DHCP Works (DORA Process)

DHCP follows a four-step process called **DORA**:

1. **Discover**
    - The client broadcasts a DHCP Discover message to find a DHCP server.
2. **Offer**
    - The DHCP server responds with an available IP address and network configuration.
3. **Request**
    - The client requests to use the offered IP address.
4. **Acknowledge (ACK)**
    - The server confirms the assignment and leases the IP address to the client.

Memory Tip: DORA = Discover → Offer → Request → Acknowledge

Example

Suppose you connect your laptop to a Wi-Fi network:

- Laptop sends **DHCP Discover**
- Router (DHCP Server) sends **DHCP Offer** with IP 192.168.1.100
- Laptop sends **DHCP Request**
- Router sends **DHCP ACK**

Your laptop is now configured automatically and can access the network.

Advantages of DHCP

- Automatic IP address assignment.
- Prevents IP address conflicts.
- Simplifies network management.
- Supports devices joining and leaving the network.
- Efficient use of available IP addresses.

Disadvantages of DHCP

- If the DHCP server fails, new devices cannot obtain IP addresses.
- Less control over specific IP assignments unless reservations are configured.
- Can be a security concern if an unauthorised DHCP server is connected to the network.

Common DHCP Ports

- **UDP Port 67** – DHCP Server
- **UDP Port 68** – DHCP Client

Interview Questions

1. What does DHCP stand for?
Answer: Dynamic Host Configuration Protocol.

2. What is the main purpose of DHCP?
Answer: To automatically assign IP addresses and other network settings to devices.

3. What is the DORA process?
Answer: Discover, Offer, Request, Acknowledge.

4. Which protocol does DHCP use?
Answer: UDP.

5. Which ports are used by DHCP?
Answer: UDP Port 67 (Server) and UDP Port 68 (Client).

Summary

- **DHCP** = Automatically assigns IP addresses.
- **DORA** = Discover → Offer → Request → Acknowledge.
- **Uses UDP** ports **67** and **68**.
- Provides **IP address, subnet mask, default gateway, DNS server, and lease time**.
- Reduces manual configuration and prevents IP conflicts.